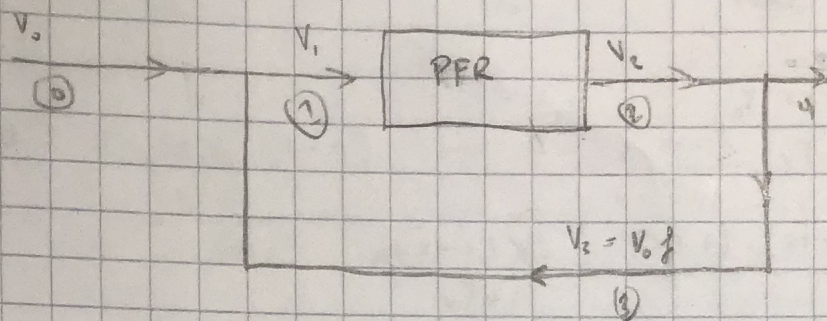


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Relationships

$$X_T = 1 - \frac{C_{A1}}{C_{A0}}$$

$$F_{A0} + F_{A3} = F_{A1}$$

$$(1) C_{A0} \rightarrow C_{A1}$$

$$C_{A3} = C_{A2}$$

$$(2) X = ?$$

$$C_{A2} = C_{A3} = C_{A4}$$

$$(I) C_{A0}V_0 + C_{A3}V_3 = C_{A1}V_1$$

$$[C_{A1}(1-x)][V_0 f] = C_{A1}V_0(1+f)$$

$$F_A = F_{A1}(1-x)$$

MASS BALANCE (PFR)

$$II \quad r_A = -KC_A$$

$$X_{PFR} = \frac{dF_A}{dV} = r_A$$

$$III \quad C_{A2} = C_{A1}(1-x)$$

II and III

$$r_A = -KC_{A1}(1-x) \rightarrow -KC_{A0}(1-x)$$

$$\frac{dF_{A1}(1-x)}{dV} = -KC_{A0}(1-x)[A]$$

$$C_{A0}V_0 + [C_{A1}(1-x)]V_0 f = C_{A1}V_0(1+f) \leftarrow$$

$$C_{A0} = (C_{A1}(1-x))f = C_{A1}(1+f)$$

$$C_{A2} = C_{A1}(-(\frac{f}{2} - fX) + (1 + \frac{f}{2}))$$

$$C_{A1} = \frac{C_{A0}V_0}{1+fX}$$

$$\frac{C_{A0}}{(-\frac{f}{2} + fX + 1 + \frac{f}{2})} = C_{A1}$$

$$\frac{d}{dv} \left[(1-x) \frac{C_{AO}}{1+f_x} V_0 (1+f) \right] = -K(1-x) \frac{C_{AO} V_0}{1+f_x}$$

$$V_0(1+f) \frac{d}{dv} \left[\frac{(1-x)}{(1+f_x)} \right] = -\frac{K(1-x)V_0}{1+f_x}$$

$$V_0(1+f) \frac{\frac{d}{dv}(1-x) \cdot 1 + dx - \frac{d}{dv}(1+f_x) \cdot (1-x)}{(1+f_x)^2} = -\frac{K(1-x)V_0}{1+f_x}$$

$$-V_0(1+f) \frac{dx}{dv} \frac{[1 + \cancel{f_x} + f - \cancel{x f}]}{(1+f_x)^2} = -\frac{K(1-x)}{1+f_x}$$

$$-V_0(1+f) \frac{dx}{dv} \frac{(1+f)}{(1+f_x)^2} = -\frac{K(1-x)}{1+f_x}$$

$$\frac{dx}{dv} \frac{(1+f)}{(1+f_x)^2} = \frac{-K(1-x)}{1+f_x} \cdot \frac{(1+f_x)}{V_0(1+f)}$$

$$\frac{dx}{dv} = \frac{-K(1-x)(1+f_x)}{V_0(1+f)^2}$$

$$\int \frac{x}{(1-x)(1+f_x)} dx = \int \frac{-K}{V_0(1+f)^2} dv$$

$$-K \int \frac{1}{1+f_x - x - f_x^2} dx = -K \int \frac{1}{V_0 + 2fV_0 + V_0 f^2} dv$$

$$\frac{\ln|f_x+1| - \ln|1-x-1|}{f+1} = -\frac{\ln(V_0)K}{(f+1)^2}$$

$$\ln|f_x+1| - \ln|1-x-1| = -\frac{\ln(V_0)K}{(f+1)^2} (f+1)$$

$$|f_x+1| - (1-x-1) = e^{\left[-\frac{\ln(V_0)K}{(f+1)} \right]}$$

$$f_x + 1 - x - 1 = 1$$

$$x = \frac{e^{\left[-\frac{\ln(V_0)K}{(f+1)} \right]}}{(f+1)} \rightarrow x = \frac{[Z]}{(f+1)}$$

$$[Z] = -\frac{\ln(V_0)K}{f+1}$$

$$1 - \frac{C_{A4}}{C_{A0}} \rightarrow 1 - \frac{C_{A2}}{C_{A0}} \rightarrow 1 - \frac{C_{A1}(1-x)}{C_{A0}}$$

$$\rightarrow \frac{1 - \frac{C_{A0}V_0(1-x)}{1+fx}}{C_{A0}} \rightarrow 1 - \frac{V_0(1-x)}{1+fx}$$

$$C_{A1} = \frac{C_{A0}}{1+fx}$$

$$\rightarrow 1 - \left[\frac{V_0 \left(1 - \frac{e^{fz}}{(f-1)} \right)}{1 + f \left(\frac{e^{fz}}{(f-1)} \right)} \right]$$